Report on Practical 7

L3 Switch Configure

1. Aims

- Understand and implement the Layer 3 switch configuration using the Cisco Catalyst 3650

2. Objectives

- To configure a Layer 3 switch for inter-VLAN routing as per the video tutorial.
- Gain practical knowledge of the role of Layer 3 switches in a network.
- Understand the fundamental differences between Layer 2 and Layer 3 switching.
- To validate network connectivity and performance through ping tests.

3. Background

- **OSI** (The Open Systems Interconnection) model is a conceptual framework that divides network communications functions into seven layers.
- A **switch** and a **router** are critical devices in a network but serve distinct purposes:

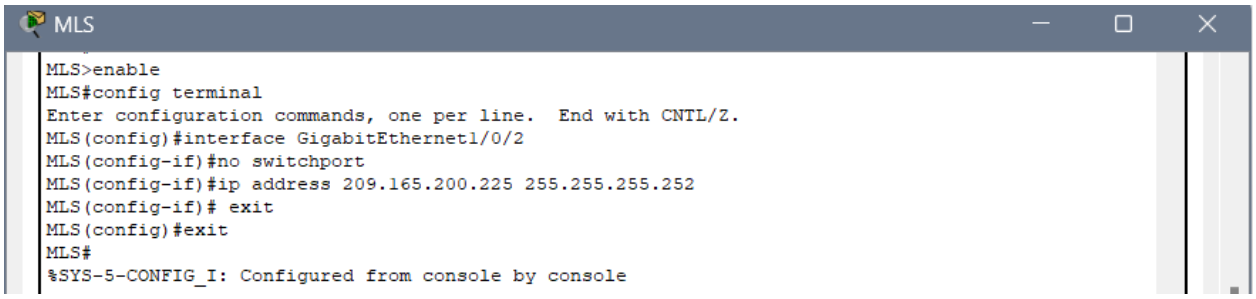
  - Switch: Operates mainly at Layer 2 of the OSI model. It connects devices in a local network (LAN) and forwards packets based on MAC addresses.

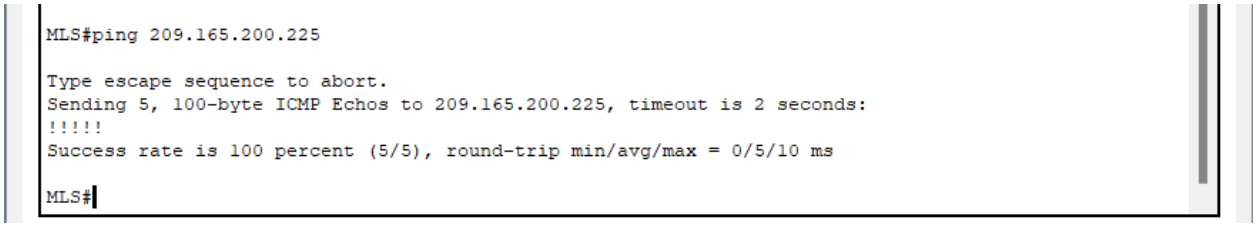
  - Router: Operates at Layer 3 of the OSI model. It connects multiple networks and forwards packets based on IP addresses.

- **Layer 2** switches are most common type. When a device connects to a layer 2 switch, it stores MAC address in its table.
- It functions as a bridge, building tables for the transfer of frames between networks, and emerged to address contention issues in shared media LANs.
- **Layer 3** switches use VLAN and separates network traffic so devices in one VLAN can't interfere with the device in other VLAN
- Layer 3 uses **Inter-VLAN routing** to allow communications between VLANs. They create **SVIs** which allow data to be routed between VLANs.

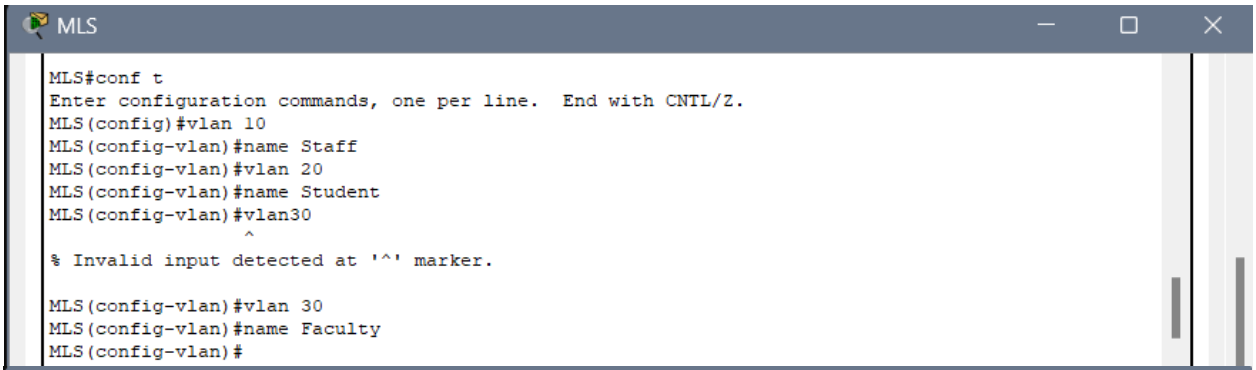
4. Procedure

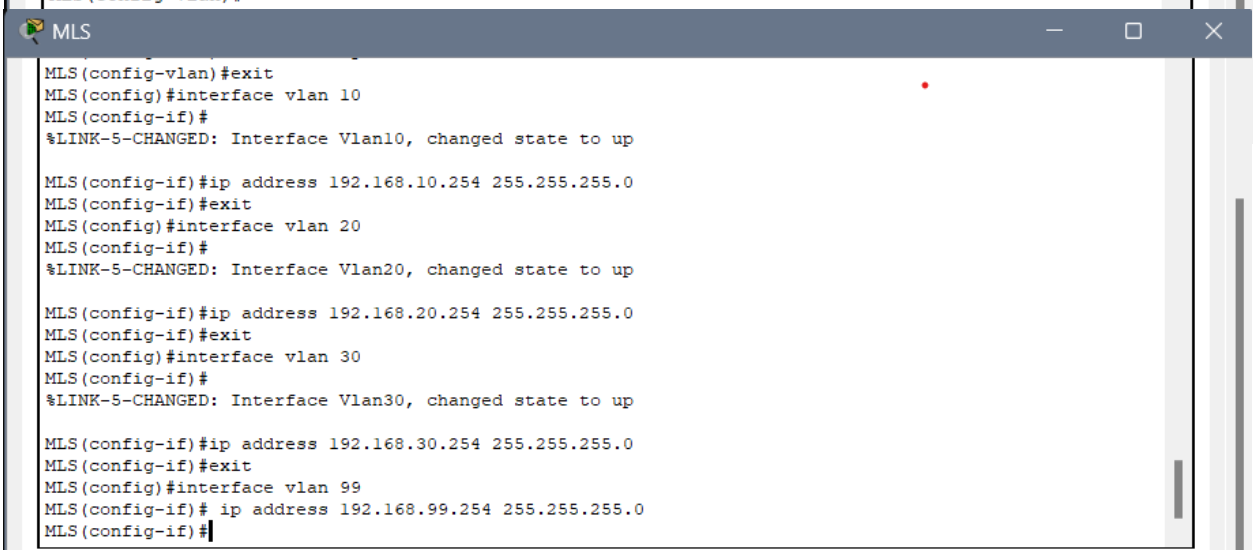
1. Configure the GigabitEthernet1/0/2 port on switch MLS as a routed port and check if we can ping



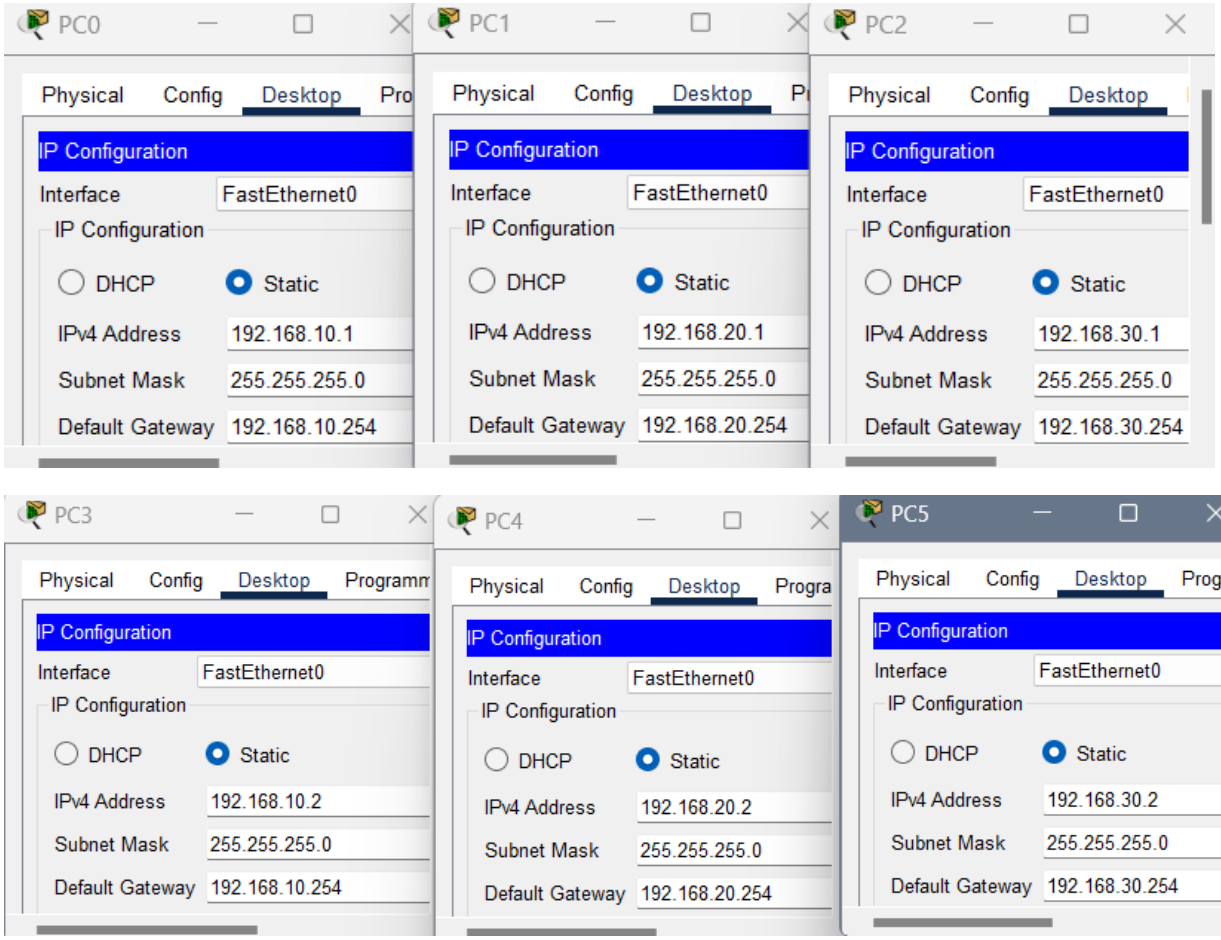


2. Add VLANs to MLS accordingly. Then Configure and activate the SVI interfaces for VLANs 10, 20, 30, and 99

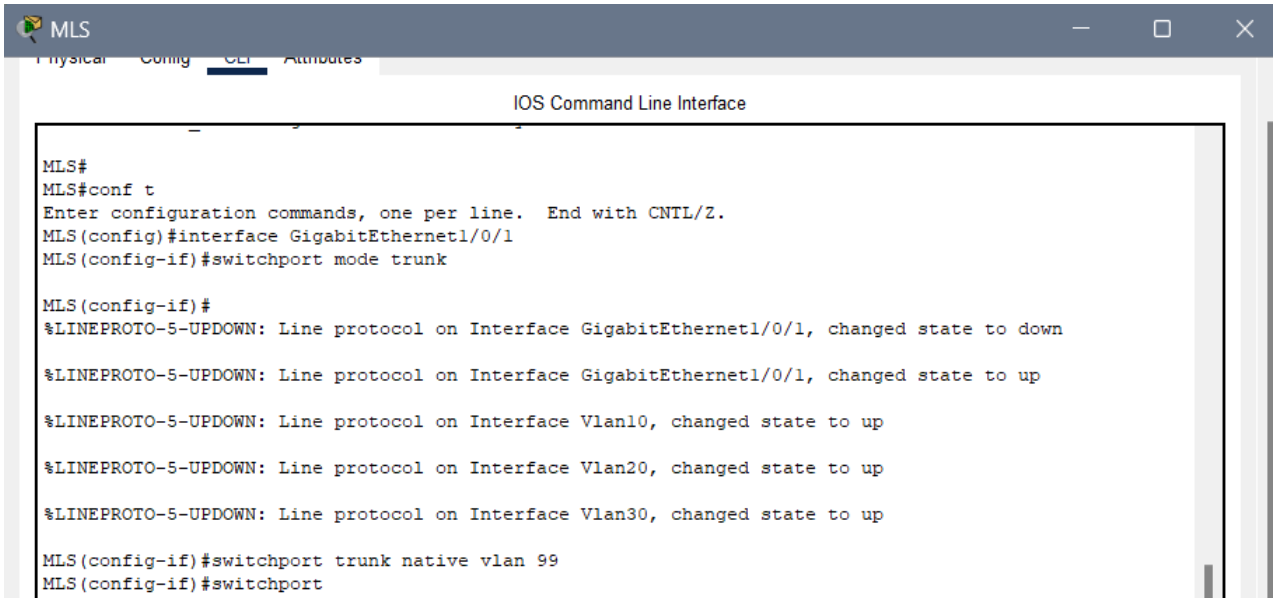




3. Assign IP address to all the PCs and Switch



4. Configure Trunking on MLS and switch1



```
Switch>enable
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface GigabitEthernet1/0/1
                                ^
% Invalid input detected at '^' marker.

Switch(config)#interface FastEthernet0/1
Switch(config-if)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (1), with MLS
GigabitEthernet1/0/1 (99).
switchport trunk native vlan 99
Switch(config-if)#
```

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5. Enable routing

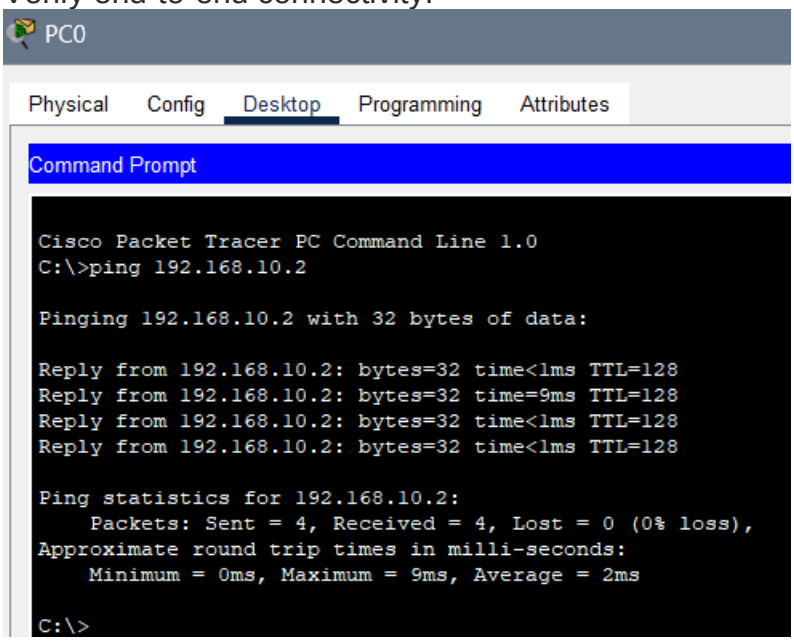
```
MLS#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
MLS(config)#ip routing
MLS(config)#exit
MLS#
%SYS-5-CONFIG_I: Configured from console by console

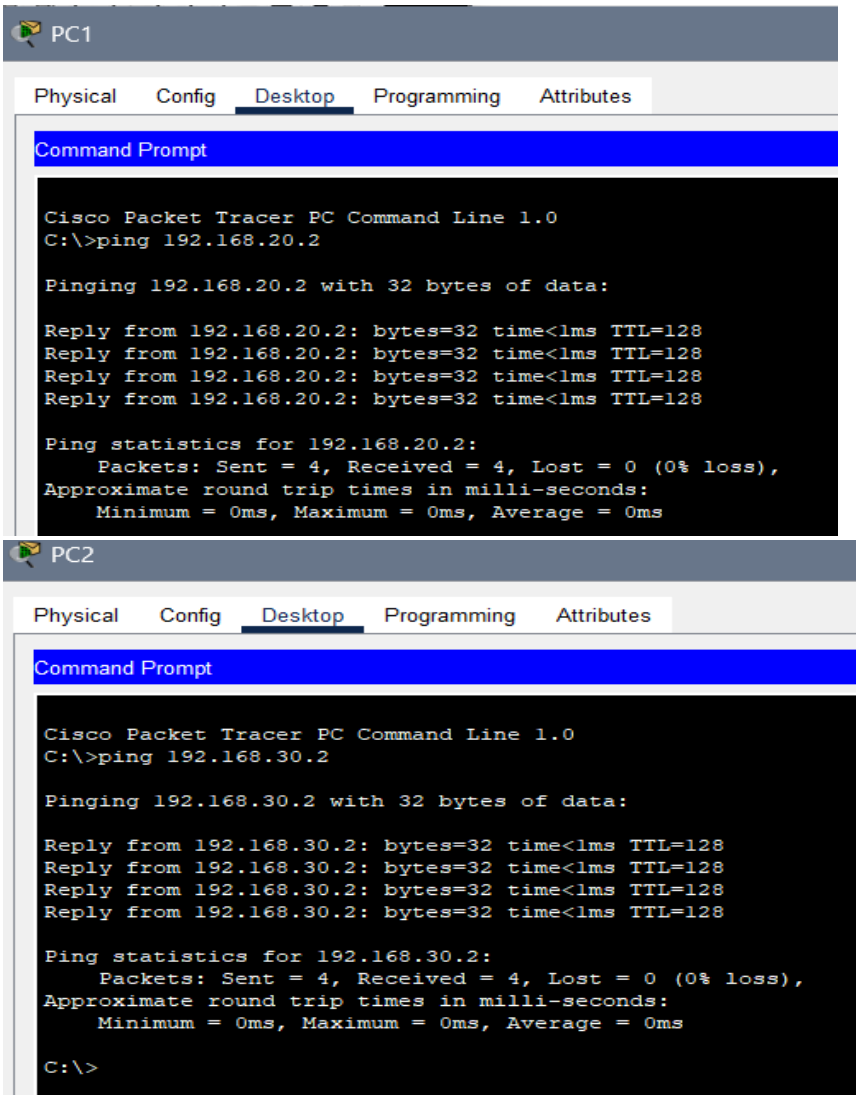
MLS#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

C    192.168.10.0/24 is directly connected, Vlan10
C    192.168.20.0/24 is directly connected, Vlan20
C    192.168.30.0/24 is directly connected, Vlan30
     209.165.200.0/30 is subnetted, 1 subnets
C       209.165.200.224 is directly connected, GigabitEthernet1/0/2
```

6. Verify end-to-end connectivity.





7. Enable IPv6 routing and configure IPv6 addressing on SVI for VLANs 10, 20, and 30

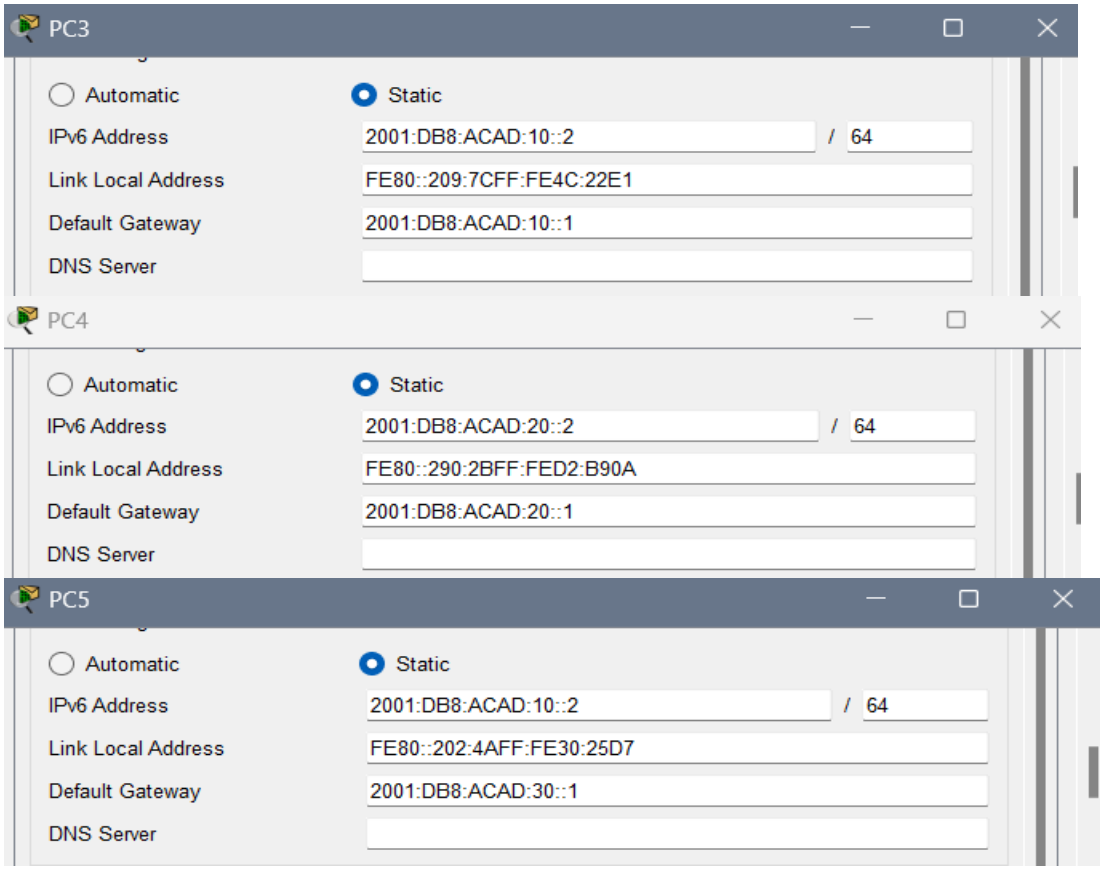
```
MLS#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
MLS(config)#ipv6 unicast-routing
MLS(config)#interface vlan 10
MLS(config-if)#ipv6 address 2001:db8:acad:10::1/64
MLS(config-if)#exit
MLS(config)#interface vlan 20
MLS(config-if)#ipv6 2001:db8:acad:20::1/64
                    ^
% Invalid input detected at '^' marker.

MLS(config-if)#ipv6 address 2001:db8:acad:20::1/64
MLS(config-if)#exit
MLS(config)#interface vlan 30
MLS(config-if)#ipv6 address 2001:db8:acad:30::1/64
MLS(config-if)#exit
MLS(config)#int GigabitEthernet1/0/2
MLS(config-if)#ipv6 address 2001:db8:acad:a::1/64
MLS(config-if)#
```

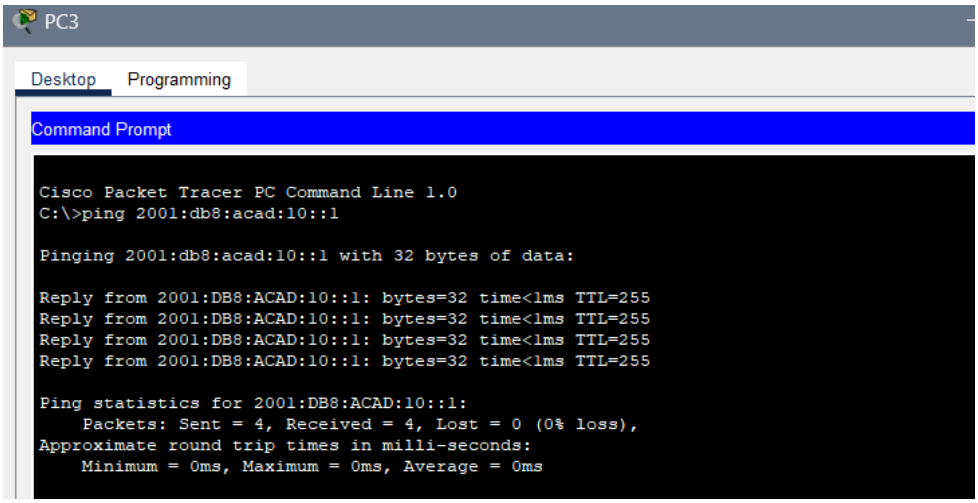
8. Use the **show ipv6 route** command to verify IPv6 connected networks.

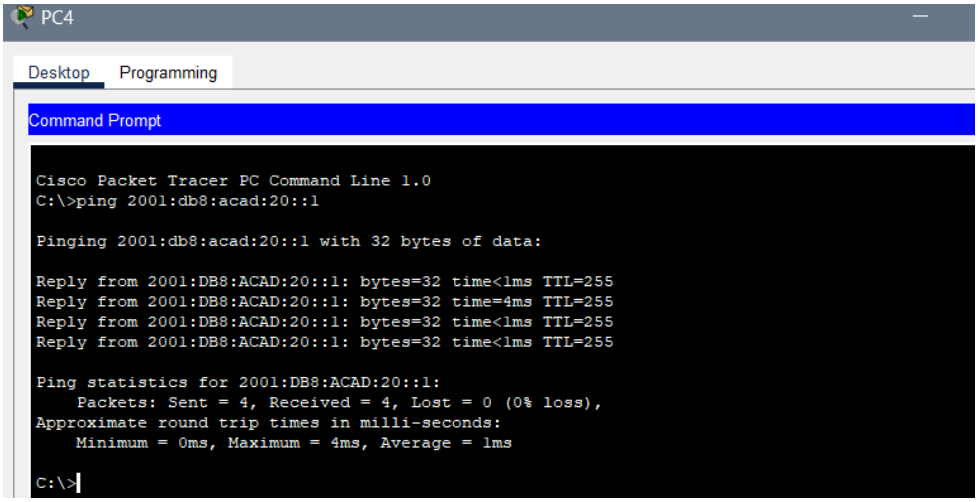
```
MLS#show ipv6 route
IPv6 Routing Table - 9 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
       U - Per-user Static route, M - MIPv6
       I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       D - EIGRP, EX - EIGRP external
C   2001:DB8:ACAD:A::/64 [0/0]
     via ::, GigabitEthernet1/0/2
L   2001:DB8:ACAD:A::1/128 [0/0]
     via ::, GigabitEthernet1/0/2
C   2001:DB8:ACAD:10::/64 [0/0]
     via ::, Vlan10
L   2001:DB8:ACAD:10::1/128 [0/0]
     via ::, Vlan10
C   2001:DB8:ACAD:20::/64 [0/0]
     via ::, Vlan20
L   2001:DB8:ACAD:20::1/128 [0/0]
     via ::, Vlan20
C   2001:DB8:ACAD:30::/64 [0/0]
     via ::, Vlan30
L   2001:DB8:ACAD:30::1/128 [0/0]
     via ::, Vlan30
L   FF00::/8 [0/0]
     via ::, Null0
MLS#
```

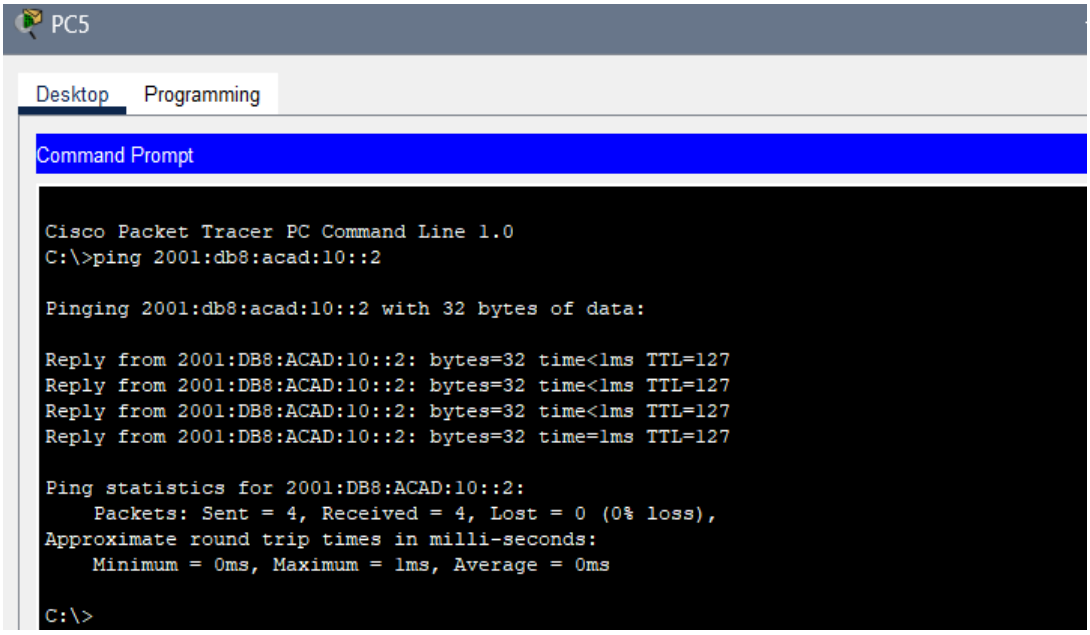
9. Configure IPv6 address to the PC3, 4 and 5

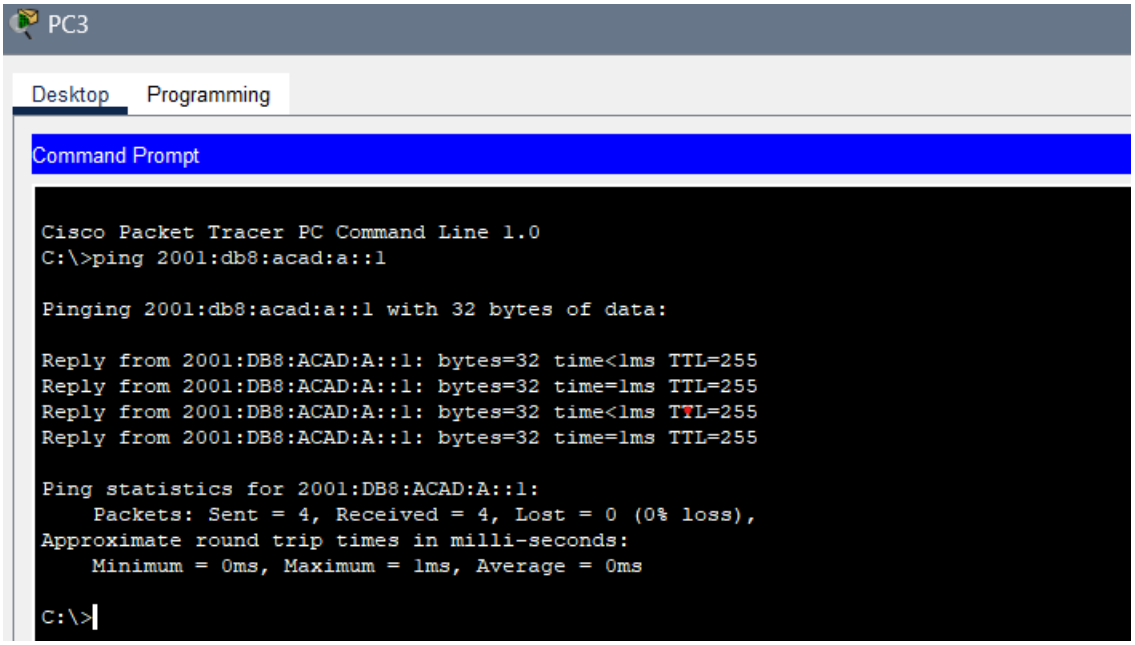


10. Verify IPv6 inter-VLAN routing and connectivity to **Cloud.**









5. Question and Answer

Q1. What is the difference between the functionality of a switch and a router?

Switch:

- Operates at Layer 2 (Data Link Layer) of the OSI model.
- Uses MAC addresses to forward frames within a local network.
- Cannot connect different networks or perform routing.

Router:

- Operates at Layer 3 (Network Layer) of the OSI model.
- Uses IP addresses to forward packets between different networks.
- Performs more advanced functions, such as Network Address Translation (NAT) and firewall filtering.

Q2: What is the difference between the working of an L2 and L3 switch?

Layer 2 Switch	Layer 3 Switch
• Works at the Data Link Layer. • Forwards data based on MAC addresses. • Does not support inter-VLAN routing. • Relatively inexpensive.	• Works at both Data Link (Layer 2) and Network Layer (Layer 3). • Routes data based on IP addresses and forwards data based on MAC addresses. • Supports inter-VLAN routing, enabling communication between different VLANs. • Significantly more expensive.

6. Conclusion

By completing the Layer 3 switch configuration Inter-VLAN communication was successfully implemented and tested. The functionality and advantages of Layer 3 switches, especially their routing capability within a single device, were also demonstrated in the video. The differences between switches, routers, and Layer 2 and Layer 3 switching were also analysed in detail. This practical highlights the importance of Layer 3 switches in modern network designs, providing both performance and versatility for complex networking requirements.

7. References

- YouTube video:

  Packet Tracer – Configure Layer 3 Switching and Inter VLAN Routing
  https://www.youtube.com/watch?v=BWhqyJYVLmI&list=PPSV&t=191s

  Layer 2 vs Layer 3 Switches

  https://www.youtube.com/watch?v=bdNS0K4Bt8U